

Market System - Status and Pre-Research Review

Version 5 | R.9I | Owner review copy | 4 September 2026

The decision today

Review the project summary and six open design decisions. No database access or implementation is requested by this report. The recommended next task, if separately authorized, is a bounded synthetic-only design investigation.

PDF_V5_GENERATED_AWAITING_OWNER_REVIEW

The project is an analysis-only Indian market research system. Eventually it should provide evidence-backed market-sentiment and stock scores, reproducible supporting evidence and visible uncertainty. Unvalidated scores remain hidden. Deterministic code is the quantitative source of truth; AI may explain evidence, not manufacture it.

Use free, lawfully accessible sources and Kite current-market data where appropriate. Never execute trades. Keep version2.0 separate: its exploratory tools and results are not canonical research evidence in custom_terminal.

Current position

- Local F&O database: **LOCATED_AND_SAMPLED_NOT_QUALIFIED.**
- Production boundary: **R9D_EXACT_PRODUCTION_ACTIVATION_IMPOSSIBLE.**
- R.9H: non-executable proposal prepared; implementation not authorized.
- PDF v4: preserved byte-for-byte, historically reviewed, now stale after R.9H.
- This PDF: report generation only; no owner answers recorded.

No connection, SQL, schema inspection, market-row access, audit, acquisition, market analysis, scoring, simulation, backtesting, recommendation, broker action or trading is authorized. No implementation or execution authority is granted.

Source checkpoint: d1f1995d92acd55318310812fa24889a30367f51.

1. What has been done - and what it proves

Implemented infrastructure

Vertical Slice A moved the existing 12-1 momentum experiment through versioned temporal data, historical universe, feature, outcome, cost, experiment and walk-forward contracts. Golden fixtures and immutable manifests support reproducibility and compatibility checks. Prediction quality, economic outcomes and portfolio behavior are separate. The result remains non-actionable, not an approved edge or production score.

Slices A.5-A.10 added data-trust assessment, provider-neutral ingestion, typed normalization, immutable raw evidence, identity reconciliation and official-source qualification. Broad local equity history is survivor-selected; public historical-population evidence remains incomplete. The official-response gate remains `AWAITING_SUBSTANTIVE_OFFICIAL_RESPONSE`. Technical access is not permission or completeness.

Kite authentication, current inventory, bounded read-only quotes and ephemeral coverage health are implemented for current-market use. Current instruments are not a historical universe. They cannot replace missing historical security snapshots or change historical trust verdicts.

Research reconciliation distinguishes exploratory, legacy and canonical evidence. Research-family, input, split-access, manifest, review and lifecycle controls support reproducibility. This machinery does not itself validate any economic hypothesis.

Synthetic evidence, not production observations

Offline governance and F&O catalog-audit tests exercise approvals, one-use consumption, failure handling and bounded metadata access on synthetic inputs. A synthetic canary tests governance, not market prediction. These tests do not establish production database safety, completeness or row accuracy.

Production observations and proposal work

R.9D deliberately disabled the production audit boundary. R.9F performed one separately authorized filesystem-only binding and bounded identity sample; it did not connect to SQLite. PDF v4 was then reviewed, authorizing proposal preparation only. R.9H delivered ten non-executable design objects covering bindings, permissions, failure states, SQLite controls, threats, tests and six open decisions.

The v4 review is historical evidence, not permission to implement R.9H. Its original generation fingerprint covered 242 files; the current post-R.9H fingerprint covers 252. No approval carries forward silently.

2. What we actually know about the F&O database

LOCATED_AND_SAMPLED_NOT_QUALIFIED means precisely that a filesystem target was found and sampled. The sanitized reference is `PRIVATE_FNO_DATABASE_V1`. Private locations and runtime binding contents are not reproduced here and were not read to prepare this report.

R.9F recorded observation	Meaning
48,345,137,152-byte file	Recorded file size, not data coverage
First 100 bytes checked	SQLite magic present; not a corruption test
64 deterministic positions	One bounded sample pass
268,435,456 sampled chunk bytes	Not a full-file hash
268,435,556 total raw bytes	Historical R.9F observation; not repeated in R.9I
Before/after checks passed	Checked metadata and sample were stable during that pass
Production SQLite connections	None
Schema and market rows inspected	None

A sampled fingerprint is not a full-file hash. Unsampled changes can be missed. Modification time is metadata, not complete identity. R.9F did not qualify sidecars, provenance, corrections, economic coverage or market-data quality. No data-quality conclusion has been established.

What the first proposed audit would do

1. Stage 1: separately authorized identity verification, including the declared header/bounded identity checks and already-local backup/retention metadata. The older scope says locate and freeze; R.9F has already bound the target. A future verification needs its own explicit authorization, not a silent repeat.
2. Stage 2: one bounded mode=ro SQLite connection, verified query_only, timeout/progress controls and enumerated catalog/table/index/foreign-key metadata. Exact optional EXPLAIN QUERY PLAN templates remain undecided. No user rows, coverage queries, quick_check or integrity_check.
3. Stage 3: bounded already-local provenance, configuration/documentation and ingestion metadata, retained raw filenames and metadata, hashes, timestamps, parser versions, corrections, vintages, backups, retention and rights evidence. Do not open market observations.

Passing Stages 1-3 would not prove market-row accuracy, historical completeness, point-in-time fitness, survivorship safety or suitability for backtesting. Stage 3 does not authorize research or qualify the entire database. Stages 4-6 remain excluded; discovery cannot expand scope.

3. Proposed one-use permission - not implemented

The proposed permission would bind one reviewed source commit, research fingerprint, current PDF and owner-review record, canonical audit specification, input declarations, dataset identity, sidecar policy, resource envelope, exact allowed statements and output contract. It would be short-lived and name exactly one attempt.

The proposed sequence is:

1. Validate exact bindings and unused, unexpired permission.
2. Atomically and durably consume permission before the first target connection.
3. Open at most one read-only connection and perform only approved catalog work.
4. Close on every path and record completion, failure or abort.

Registration is not consumption. Once consumed, permission cannot be replayed or automatically retried, even after a crash before connection. A missing terminal event remains detectable as an incomplete consumed attempt. Every retry would require a new exact approval. Mismatch or mutation must fail closed, not trigger repair or scope expansion.

Separate gates

Proposal review is not implementation approval. Implementation approval is not audit-execution approval. Audit approval is not market-research approval. No scoring, analysis, simulation, backtesting, recommendation or trading authority is included.

The proposed protections layer mode=ro, verified query_only, an authorizer, disabled extensions, exact statement templates, escaped catalog identifiers, counters, deadlines and identity checks. Mode=ro alone does not enforce identity, permissions, sidecar consistency, bounded I/O or safe scope.

Limits must not be oversold

Limit category	Current meaning
Code-enforced in tested synthetic paths	Selected statement, result and output guards are exercised offline; this is not production proof.
Monitored or checked	Deadlines and before/after identity checks can detect some breaches; detection is not prevention of all races or I/O.
Declared or unresolved	Production database-byte budget and whole-process memory, total wall-time and temporary-storage limits still need enforceability decisions.

The proposal declares 1 connection, 50 attempted statements, 5 seconds per statement, 1,200 seconds total, 25,000 result rows, 256 MiB database reads and 25 MiB output. These are proposed bounds, not evidence of production enforcement. SQLite progress callbacks alone cannot enforce a hard database-byte or whole-process resource limit.

4. Decision 1 - Quiet database or sidecars?

Question: Must the future audit use a quiescent database with no WAL, SHM or journal files, or explicitly bind a reviewed set of those files?

Why it matters: SQLite sidecars can contain state not represented by the main-file sample. Ignoring them can produce incomplete or inconsistent observations. An absence check alone does not prove no writer can start later.

Options and trade-offs: A no-sidecar, no-active-writer policy is simpler and easier to review, but may reject a legitimate operational database. Binding a sidecar set may support more states, but requires a reviewed consistency protocol, identity coverage and mutation handling for every relevant object.

PROPOSED_NOT_APPROVED: Prefer the simplest first-audit policy: require demonstrated quiescence and no WAL/SHM/journal sidecars; abort if that cannot be established or changes. Do not delete, checkpoint, repair or change journal mode to make the target qualify. This is a proposed prerequisite, not a claim that the current database satisfies it.

Engineering evidence needed: Synthetic tests for concurrent writers, sidecar appearance, path substitution and mutation races; a credible method to establish and maintain quiescence on the target OS. If that cannot be demonstrated, keep access blocked and review a separate consistency design.

Gate: Blocks boundary implementation design and later execution. No private-file probing is authorized to resolve it now.

5. Decision 2 - What does a read budget measure?

Question: How can target database bytes read be capped, or should the declared byte cap be replaced with page and statement budgets with explicit limitations?

Why it matters: Bytes read, pages visited, statements executed and returned rows measure different work. Revisited/cached pages, read-ahead, page sizes and filesystem caching prevent treating them as interchangeable. Few returned rows can still require substantial reads.

Options and trade-offs: Prove enforceable byte accounting at an appropriate I/O boundary, with a defined metric; or separately revise the contract to narrower measurable limits and explicitly accept what they do not bound. Neither mechanism is proven by the current proposal.

PROPOSED_NOT_APPROVED: Retain this as an implementation blocker. Authorize only a later bounded synthetic investigation comparing mechanisms; do not relabel rows, statement counts or progress callbacks as a 256 MiB byte cap. Do not silently substitute weaker budgets.

Engineering evidence needed: Define whether the cap counts logical database I/O or physical storage reads, sidecar inclusion, cache behavior, cancellation and overshoot. Test actual enforcement and failures with synthetic databases on the intended runtime. Any replacement requires a separately reviewed versioned specification.

Gate: Blocks production boundary implementation and execution until an honest, testable resource contract is resolved.

6. Decision 3 - Explicit or derived attempt ID?

Question: Should permission name one attempt directly, or derive its identifier from the sealed approval and an exact operator nonce?

Why it matters: A unique attempt links consumed permission to one terminal result. Ambiguous identifiers can complicate replay rejection, concurrent runs and crash recovery.

Options and trade-offs: An explicit identifier is easier for an owner to inspect but needs uniqueness checks. A derived identifier is reproducible but requires an exact canonicalization rule, nonce handling and no circular dependency between approval hash and attempt ID.

PROPOSED_NOT_APPROVED: Prefer an explicitly named attempt ID included in the sealed payload, with durable uniqueness checks. This reduces review complexity; it is not yet an approved design.

Engineering evidence needed: Synthetic duplicate/concurrency/crash tests, payload mutation rejection and evidence that the approved ID is the only ID accepted. If derivation is preferred, specify the exact non-circular hash inputs and nonce lifecycle before implementation.

Gate: Blocks final implementation contract; later audit approval must bind the resolved exact attempt.

7. Decision 4 - Is a query plan needed?

Question: Which exact EXPLAIN QUERY PLAN templates, if any, are needed for Stages 1-3?

Why it matters: Catalog inspection asks about structure and safety, not market coverage. Planning later analytical queries can unnecessarily broaden the first audit's surface even without executing those queries.

Options and trade-offs: Use no plan templates for the initial metadata-only audit; or justify a minimal exact, hash-bound set needed for an explicit Stage 1-3 question. No caller-supplied SQL or explained-statement execution is acceptable.

PROPOSED_NOT_APPROVED: Prefer no EXPLAIN QUERY PLAN templates unless a specific metadata-stage need is demonstrated. The canonical scope currently permits planning later queries; this recommendation does not edit that scope or add queries. Resolve any formal narrowing in a separately reviewed specification.

Engineering evidence needed: Map every proposed statement to a Stage 1-3 question, prove metadata output is sufficient where claimed and test rejection of all other templates using synthetic fixtures only.

Gate: Blocks final query allowlist before implementation; every later execution must use the reviewed list without discovery-driven expansion.

8. Decision 5 - Declared versus enforced resources

Question: Are declared but unenforced process memory, whole-attempt wall-time and temporary-storage limits acceptable for the first audit?

Why it matters: A limit written in JSON is not a safeguard by itself. A per-statement progress handler does not constrain all process activity, blocking calls, memory allocations or temporary files. Monitoring may detect a breach only after it occurs.

Options and trade-offs: Prove enforcement using a separately designed supervisor/OS mechanism; or explicitly accept residual risk under a revised contract with monitored/declared bounds identified. A simpler design may be less protective. No claim of hard enforcement is justified by current evidence.

PROPOSED_NOT_APPROVED: Do not accept unspecified unenforced limits as sufficient now. First run a bounded synthetic-only feasibility investigation of enforceable process limits and safe terminal recording. Report unavoidable overshoot and monitoring gaps before asking the owner to accept a precise limitation.

Engineering evidence needed: Tests for stalled calls, excess allocation/output/temp use, forced termination, durable consumption and missing-terminal reconciliation. Classify each control as enforced, monitored or declared; show its failure behavior on the intended OS.

Gate: Blocks completion of the implementation safety contract and production execution. It does not justify private-database experiments.

9. Decision 6 - Future one-use audit permission

Question: After implementation review, may a separate task create one exact short-lived audit approval, with no research authority?

Why it matters: Preparing safe code and approving its use are different decisions. Permission must refer to actual reviewed code, evidence and limits, not a future unknown implementation.

Options and trade-offs: Defer the decision until the required evidence exists; or reject future audit access. Approving execution now is not an eligible option in this milestone.

PROPOSED_NOT_APPROVED: Defer. After design resolution and separate implementation authorization, require successful adversarial tests, review of exact code and a current report before requesting one-use audit authority. This PDF does not ask for immediate execution approval.

Engineering evidence needed: Reviewed implementation commit, current fingerprint/report, exact inputs, sidecar and resource decisions, unusable-template rejection, crash/replay tests and operator checks. None is replaced by a generic instruction to continue.

Gate: Blocks later approval issuance and execution. Resolving design questions or authorizing an implementation is not this approval. No approval record is created by v5.

10. What happens next

1. Owner reviews PDF v5 and answers its questions. No answers are pre-filled.
2. Resolve technical design blockers through separately authorized, bounded synthetic-only work.
3. Review that evidence and only then consider a narrowly defined implementation task.
4. Implementation does not authorize a production audit. Reconcile the implemented code and tests; refresh the status report if the research state changes.
5. Before production access, require a current report and a separate exact short-lived one-use approval bound to the reviewed implementation and inputs.
6. Review any authorized Stage 1-3 results before proposing further data qualification. Passing them is not research fitness.
7. Require the owner's current pre-research PDF review and explicit scoped authority before analysis, simulation or backtesting. Scores must still meet independent data and validation gates; trading remains prohibited.

One recommended next task - not authorized here

PROPOSED_NOT_APPROVED: BOUNDED_SYNTHETIC_BOUNDARY_DESIGN_INVESTIGATION_ONLY.

Resolve sidecar/quiescence feasibility, byte-budget semantics/enforcement, process-resource enforcement and exact metadata query necessity. Compare attempt-ID alternatives on synthetic fixtures. Produce findings and a revised proposal only if separately authorized; do not connect the private database or remove the interlock.

If the local source fails

Preserve the lawful free-source fallback policy: compare alternatives, including official NSE F&O bhavcopies for all lawfully available and accessible history. The candidate is not a chosen or qualified source. Official availability alone does not prove completeness, retention rights, point-in-time fidelity, identity continuity, correction coverage or research fitness.

No acquisition or external-source access is authorized in R.9I. Do not bypass CAPTCHA, authentication, robots rules, rate limits or licences. Apply the same provider-neutral provenance, completeness, correction, retention and reproducibility gates to any alternative. Current Kite data remains current-only; version2.0 remains separate and noncanonical.

11. Owner questions - please answer by number

Summary confirmation

1. Is this project summary accurate enough to proceed with review: analysis-only Indian research, free lawful sources, eventual evidence-backed scores, unvalidated scores hidden, version2.0 separate and no trade execution?
2. Is the evidence distinction clear and accurate: synthetic tests are not production qualification; R.9F is sampled identity only; no production SQLite connection, schema or rows have been inspected? Please identify corrections or missing constraints.

Policy preferences - all recommendations are PROPOSED_NOT_APPROVED

3. Do you prefer the no-sidecar/quiescent first-audit policy, subject to engineering proof, rather than a more complex sidecar binding? It may legitimately block the target; no repair or checkpoint is implied.
4. Do you prefer an explicit attempt ID bound in the future approval, subject to uniqueness/replay tests, rather than a derived ID?
5. Do you prefer no EXPLAIN QUERY PLAN templates unless a specific Stage 1-3 need is proven and reviewed? This does not authorize any SQL or change the current scope.

Technical questions needing evidence, not an owner's guess

6. Should the byte-budget and process-limit decisions remain blocked until synthetic evidence defines what is actually enforced, monitored or merely declared? Recommended: yes; do not accept unspecified unenforced limits now. You are not being asked to certify a technical mechanism.
7. Are the sampled-identity limitations and the fact that Stages 1-3 cannot establish historical completeness or backtesting fitness understood? Accepting these facts is not accepting every unresolved safety risk.

Authorization for a specific future task only

8. Do you authorize a separate bounded synthetic-only design investigation as described on the previous page, with a report of findings but no production boundary implementation, interlock change, private database/config/binding access, approval issuance, acquisition or market research?

Deferred future gate

9. Confirm that any future one-use audit approval must be requested separately after reviewed implementation and a current report. No immediate audit authority is requested. Please add any corrections or constraints before that future request.

Silence, a general "continue", or approval of this summary is not implementation or audit authority. No answers, consent or owner-review record have been recorded for PDF v5.

Appendix - Evidence and verification boundary

The non-approval generation manifest binds the exact Markdown, generator, PDF and sanitized evidence inputs. This report summarizes source checkpoint d1f1995d92acd55318310812fa24889a30367f51; it does not claim a new research-state revision.

Evidence	Exact hash or state
Research fingerprint (252 files)	1b56c28fabed28672d140cf76ba8b242f00e0b4965ab682ebc7704bb38742fef
R.9H proposal manifest	a6529ec14520d163e327e2dcc7a7f469ea473d14b8d39e3ede345bc3d49dcdc1
R.9H package content	628be5879d34a0a2f7c53ca493279c0a797f4ad45575988083d58e7b27f2945f
R.9F anchor	115eb8da500a81455061c13c130ee458496b38190caf11dbe4bba35386652acc
R.9F binding proposal	995524b670dc95b717fa7d4b27935c788d661bcf75b8f7f4400d76831a8f434f
R.9F sampled root (not full-file hash)	b1b8c0ca1338d477987da28e6d9647b151c120a0eac7bb17c9e9293edfd4bc47
Canonical Stage 1-3 audit scope	86af643f9c88a1162ecaecf76d22e4e74b5d39816f6e6117eb6105b726588d8d
Historical PDF v4 (14 pages)	02a76f6d46bc74a69b7f0b10331ae26da1d07d60934091ce9d31c0abe8cdaec9

Source map

Read in full: R.9H's ten-object proposal package and RESEARCH_R9H_REPORT; canonical audit_scope_v1; PDF v4 Markdown, generation manifest and owner-review record. The current v4 review-record hash is bound by the v5 generation manifest. Its lifecycle changed after proposal preparation; the older hash recorded inside R.9H remains historical and is not silently replaced.

The fingerprint policy is unchanged. Report-only documents, PDF and focused tests do not add research authority or change the 252-file research state. Protected source, specifications, earlier PDFs and evidence are preserved. No private database, configured path value or runtime binding contents were read for this work.

Historical R.9H verification: root 382 passed / 3 skips / 2 warnings; separate Data test 289 passed; 101 JSON files valid. These are historical counts, not R.9I test results. Newly observed R.9I commands and results are recorded in RESEARCH_R9I_REPORT and the generation manifest, including deterministic rendering and every-page visual review.

PDF_V5_GENERATED_AWAITING_OWNER_REVIEW

All implementation and execution authority remains false. Stop for human review.